Solvent-Dependent Charge Transfer and Acidity in *p*-Nitroaniline

Course Report

Name:	Arif Ahmed Begg
Roll Number:	24CY40005
Email:	arifahmedbegg.24@kgpian.iitkgp.ac.in
Course:	Modelling of Quantum Systems & Processes
Instructor:	Prof. Sabyashachi Mishra

Department of Chemistry
Indian Institute of Technology Kharagpur

Abstract

Charge transfer (CT) transitions in donor–acceptor aromatic systems are strongly affected by the nature of the exchange–correlation functional and by solvent polarity. In the present study, aniline, nitrobenzene, and *p*-nitroaniline were used as a chemically connected series to move from a donor-substituted ring, to an acceptor-substituted ring, and finally to a para donor–acceptor push–pull chromophore. Ground-state geometries were optimized at the B3LYP/6-31G(d) level. For *p*-nitroaniline, TD-DFT calculations were carried out with B3LYP and CAM-B3LYP for five singlet excited states in the gas phase and in benzene, acetonitrile, and water. The solvent trend shows a systematic bathochromic shift, while the functional trend shows that B3LYP predicts substantially lower excitation energies than CAM-B3LYP for the same bright transition.

Introduction

Electronic absorption spectroscopy probes transitions between occupied and unoccupied molecular orbitals. The excitation energy is related to the observed wavelength through $\Delta E = h\nu = \frac{hc}{\lambda}$ and the intensity of a transition is represented by the oscillator strength. In aromatic molecules, the common low-lying excitations are local $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions. When donor and acceptor substituents are connected through a conjugated framework, an additional class of transition becomes possible, namely intramolecular charge transfer.

p-Nitroaniline combines both functionalities in a para arrangement, creating a push–pull system in which charge can be displaced from the amino donor toward the nitro acceptor through the aromatic π framework. [1, 2] This makes *p*-nitroaniline a useful model system for examining charge-transfer excitations.

The aim of this report is therefore not only to describe the calculated spectra, but also to identify which computed band corresponds to charge transfer and to show how the predicted position of this band depends on the functional and on the dielectric environment.

Computational Methodology

All ground-state geometries were optimized at the B3LYP/6-31G(d) level. Excited-state calculations were then carried out using time-dependent density functional theory, and five singlet excited states were requested in each calculation. For *p*-nitroaniline, two functionals were used in the excited-state step: B3LYP and CAM-B3LYP. The latter was included to assess the effect of long-range corrected exchange on the predicted position

of the charge-transfer band. Solvent effects were incorporated through the SCRF/PCM framework using benzene, acetonitrile, and water as media of increasing polarity.

The quantitative discussion below is based directly on the eight Gaussian TD-DFT outputs for *p*-nitroaniline: gas phase and three solvents, each evaluated with B3LYP and CAM-B3LYP. [3] All calculations used `td=(singlets,nstates=5)`. The complete TD-DFT state-by-state tables are placed in Appendix A, while the main text focuses on the states that are directly relevant to the assignment of the charge-transfer band.

Results and Discussion

Electronic Structure: Aniline, Nitrobenzene, and *p*-Nitroaniline

Aniline raises the occupied orbital manifold through resonance donation from the nitrogen lone pair into the ring, whereas nitrobenzene lowers the acceptor orbitals because of the strongly electron-withdrawing nitro group. In *p*-nitroaniline these two effects are combined in a para donor–acceptor arrangement, so the lowest optically active transition is expected to involve electron promotion from a donor-enriched occupied orbital into an acceptor-enriched virtual orbital. The TD-DFT results for *p*-nitroaniline confirm this expectation, because the bright low-energy state is repeatedly dominated by the same frontier-orbital excitation, 36→37, across gas phase and solvents.

Gas-Phase Excited States of *p*-Nitroaniline

The complete gas-phase TD-DFT state listings are provided in Appendix A. In the main text, only the spectroscopically relevant features are emphasized. The most important point is that the bright state is not the first excited state in either functional. At B3LYP, the first excited state at 324.94 nm is dark and dominated by 34→37 / 34→39 mixing, whereas the intense low-energy absorption is the second excited state at 302.72 nm with $f = 0.3226$, dominated by 36→37. At CAM-B3LYP, the first two states remain weak and the bright low-energy band appears as the third excited state at 272.97 nm with $f = 0.3798$, again dominated by 36→37.

This functional dependence is best understood in a state-specific way rather than as a simple overall shift. B3LYP places the bright 36→37 state at 302.72 nm, whereas CAM-B3LYP places the corresponding bright state at 272.97 nm. The shift of almost 30 nm toward shorter wavelength with CAM-B3LYP is consistent with the expected behaviour of a CT-sensitive transition, because long-range corrected exchange reduces the overstabilization of charge-separated excitations that is common with conventional hybrid functionals.

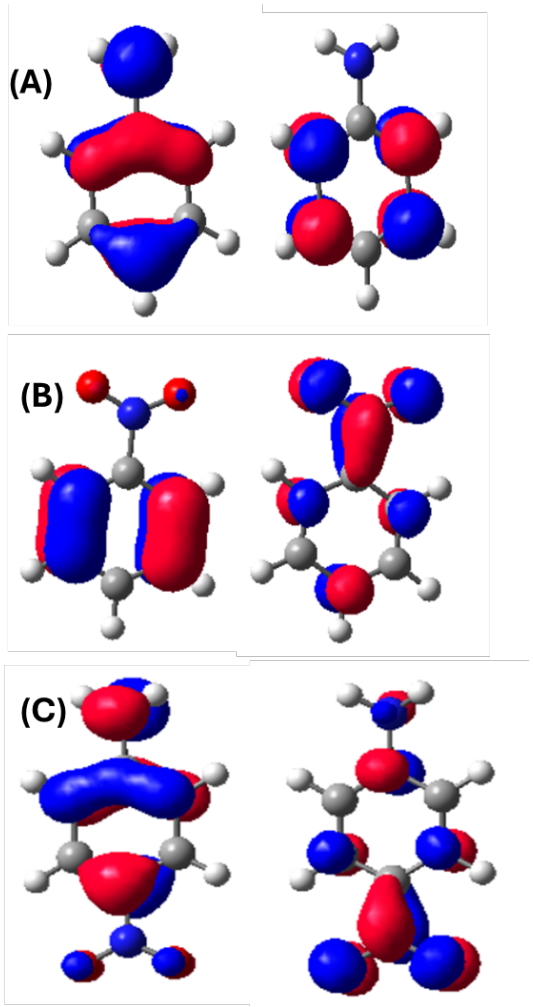

Figure 1: HOMO-LUMO of (A) Aniline (B) Nitrobenzene (C) *p*-nitro aniline

Table 1: Charge-transfer band of *p*-nitroaniline across media.

Medium	Functional	Wavelength (nm)
Gas phase	B3LYP	302.72
Gas phase	CAM-B3LYP	272.97
Benzene	B3LYP	322.66
Benzene	CAM-B3LYP	289.92
Acetonitrile	B3LYP	333.69
Acetonitrile	CAM-B3LYP	299.47
Water	B3LYP	334.10
Water	CAM-B3LYP	299.84

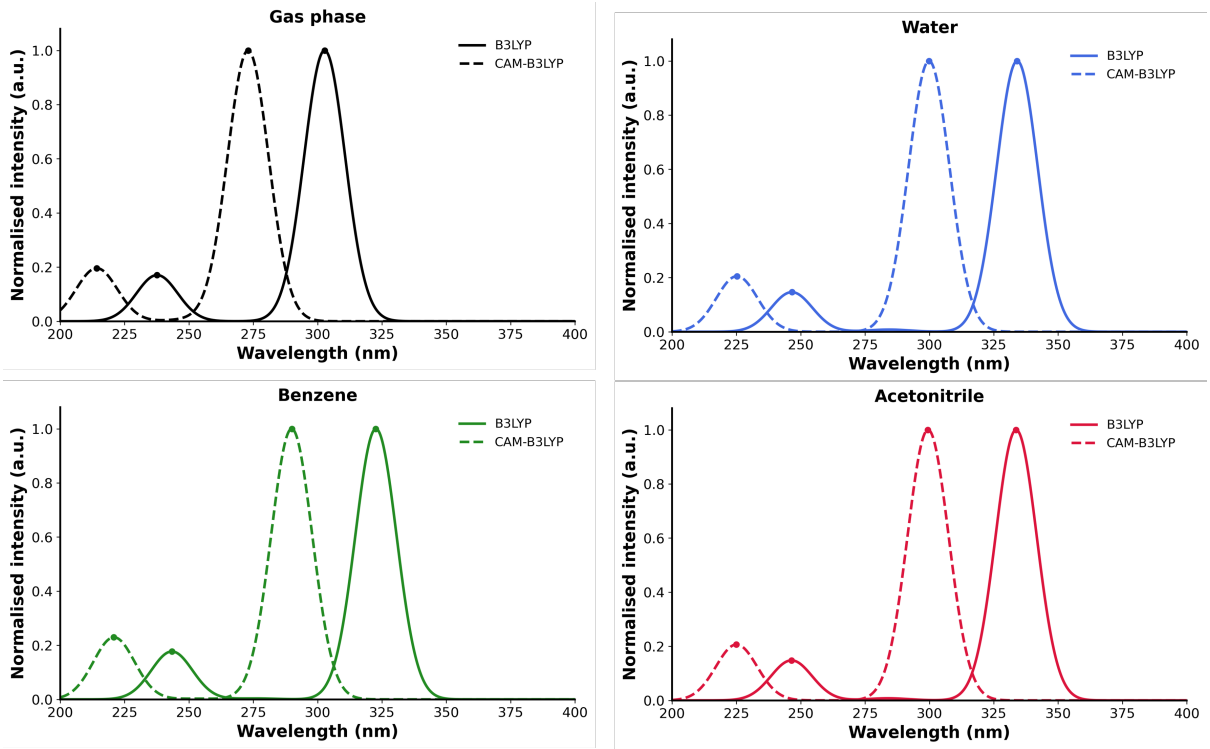

Figure 2: Computed gas-phase and solvent dependent spectra of *p*-nitroaniline showing the functional dependence

Solvent Dependence of the Excited States

The full solvent-phase TD-DFT data are collected in Appendix A. The same frontier transition remains dominant throughout, but its state ordering and wavelength depend on both the solvent and the functional.

At B3LYP, the bright CT state is already the first excited state in benzene, acetonitrile, and water, and it remains dominated by 36→37 with oscillator strengths close to 0.40. The corresponding wavelengths are 322.66 nm in benzene, 333.69 nm in acetonitrile, and 334.10 nm in water. This shows a clear bathochromic shift from gas phase to solution, with the major red shift appearing on moving from gas phase into a dielectric medium.

At CAM-B3LYP, the first excited state remains dark in all solvents, while the bright CT band appears as the second excited state. It is centred at 289.92 nm in benzene, 299.47 nm in acetonitrile, and 299.84 nm in water, each time with oscillator strength close to 0.47 and a dominant 36→37 contribution accompanied by a small admixture from 31→37.

Across gas phase and all solvents, the bright low-energy band is repeatedly dominated by the 36→37 excitation at B3LYP and in the bright CAM-B3LYP states. This persistence is characteristic of one chemically meaningful transition rather than unrelated local excitations that would reshuffle completely from medium to medium.

Excited-State Acidity Calculation

The excited-state acidity of p-nitroaniline was estimated using a Förster-type relation (see Appendix B), with TD-DFT calculations performed at the CAM-B3LYP/6-31G(d) level. The shift in acidity is given by

$$\Delta pK_a^* = pK_a^* - pK_a \approx -16.91\,\Delta E(\text{eV}) \tag{1}$$

where

$$\Delta E = E_{\text{acid}} - E_{\text{base}} \tag{2}$$

The excitation energies were obtained directly from the TD-DFT output. The values used are summarised in Table 2.

Table 2: Excitation energies of neutral (base) and protonated (acid) p-nitroaniline.

Solvent	E_{base} (eV)	E_{acid} (eV)
Benzene	4.2766	5.1595
Acetonitrile	4.1401	4.9698
Water	4.1350	4.9595

The corresponding energy differences and excited-state acidity shifts are given in Table 3.

Table 3: Calculated excited-state acidity shifts.

Solvent	ΔE (eV)	ΔpK_a^*	pK_a^*
Benzene	0.8829	-14.93	-13.93
Acetonitrile	0.8297	-14.03	-13.03
Water	0.8245	-13.94	-12.94

The positive values of ΔE in all solvents lead to large negative values of ΔpK_a^*, indicating that the molecule becomes significantly more acidic in the excited state. The magnitude of this effect is similar across solvents, with benzene showing a slightly larger shift.

This behaviour is directly related to the charge-transfer (CT) nature of the excitation. In neutral p-nitroaniline, the dominant transition involves electron transfer from the amino donor to the nitro acceptor. Upon protonation, the donor ability of the amino group is reduced, leading to suppression of the CT character. As a result, the excited-state energy of the protonated species increases relative to the neutral molecule, giving rise to large positive values of ΔE and consequently a strong decrease in pK_a^*.

Conclusion

The original purpose of this study was to understand charge transfer in a chemically connected series culminating in p-nitroaniline. The TD-DFT results show that the decisive spectral feature is the bright frontier-orbital transition dominated by 36→37. This transition defines the charge-transfer band of p-nitroaniline.

At B3LYP the CT band occurs at 302.72 nm in the gas phase and red-shifts to 322.66 nm in benzene, 333.69 nm in acetonitrile, and 334.10 nm in water. At CAM-B3LYP the corresponding bright band occurs at 272.97 nm in the gas phase and shifts to 289.92 nm in benzene, 299.47 nm in acetonitrile, and 299.84 nm in water. The CT band is therefore both solvent-sensitive and functional-sensitive, which is the main result of this work. p-nitroaniline behaves as a donor–acceptor chromophore, where the amino donor and nitro acceptor generate a low-energy optically allowed transition.

The excited-state acidity of p-nitroaniline shows a significant decrease in pK_a^* across all solvents, indicating a strong enhancement of acidity upon excitation. This behaviour arises from the suppression of donor-to-acceptor charge-transfer character upon protonation.

References

[1] Jacopo Tomasi, Benedetta Mennucci, and Roberto Cammi. Quantum mechanical continuum solvation models. *Chem. Rev.*, 105:2999–3094, 2005.

[2] Takeshi Yanai, David P. Tew, and Nicholas C. Handy. A new hybrid exchange–correlation functional using the coulomb-attenuating method (cam-b3lyp). *Chem. Phy. Lett.*, 393:51–57, 2004.

[3] Mark E. Casida. Time-dependent density functional response theory for molecules. *Rec. Adv. in DFT Methods*, 1:155–192, 1995.

Appendices

A TD-DFT Excited-State Tables

Table 4: Gas-phase TD-DFT excited states of *p*-nitroaniline at B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	3.8156	324.94	0.0000	34→37 (0.69760) 34→39 (-0.11317)
S_2	4.0957	302.72	0.3226	36→37 (0.69885)
S_3	4.3593	284.41	0.0001	33→37 (0.69804) 33→39 (-0.11004)
S_4	4.6165	268.57	0.0000	35→37 (0.59474) 36→38 (-0.36453)
S_5	5.2178	237.62	0.0550	35→37 (0.36541) 35→39 (-0.16447) 36→38 (0.57874)

Table 5: Gas-phase TD-DFT excited states of *p*-nitroaniline at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	3.9633	312.83	0.0000	34→37 (0.67690) 34→39 (-0.19947)
S_2	4.4898	276.15	0.0003	32→37 (0.67782) 32→39 (-0.19337)
S_3	4.5421	272.97	0.3798	36→37 (0.69093)
S_4	4.9545	250.25	0.0025	35→37 (0.46825) 35→39 (0.12456) 36→38 (0.50840)
S_5	5.7891	214.17	0.0745	35→37 (0.52592) 36→38 (-0.45744)

Table 6: TD-DFT excited states of *p*-nitroaniline in benzene at B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	3.8426	322.66	0.4135	36→37 (0.70284)
S_2	3.8872	318.96	0.0000	34→37 (0.69918) 34→39 (-0.10466)
S_3	4.4214	280.42	0.0001	33→37 (0.69926) 33→39 (-0.10212)
S_4	4.5018	275.41	0.0014	35→37 (0.64141) 36→38 (-0.28611)
S_5	5.0946	243.37	0.0732	35→37 (0.28570) 35→39 (-0.15391) 36→38 (0.62638)

Table 7: TD-DFT excited states of *p*-nitroaniline in benzene at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	4.0254	308.00	0.0000	34→37 (0.68090) 34→39 (-0.18633)
S_2	4.2766	289.92	0.4803	31→37 (-0.10264) 36→37 (0.69357)
S_3	4.5416	273.00	0.0001	32→37 (0.68159) 32→39 (-0.18080)
S_4	4.8918	253.45	0.0013	35→37 (0.50772) 35→39 (0.11795) 36→38 (-0.47316)
S_5	5.6134	220.87	0.1102	35→37 (0.48876) 36→38 (0.50169)

Table 8: TD-DFT excited states of *p*-nitroaniline in acetonitrile at B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	3.7156	333.69	0.4043	36→37 (0.70359)
S_2	3.9692	312.36	0.0000	34→37 (0.70041)
S_3	4.3657	283.99	0.0031	35→37 (0.67121) 36→38 (-0.21276)
S_4	4.4953	275.81	0.0001	33→37 (0.70030)
S_5	5.0317	246.41	0.0598	35→37 (0.21032) 35→39 (-0.15645) 36→38 (0.65522)

Table 9: TD-DFT excited states of *p*-nitroaniline in acetonitrile at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	4.0979	302.55	0.0000	34→37 (0.68434) 34→39 (-0.17309)
S_2	4.1401	299.47	0.4677	31→37 (-0.10474) 36→37 (0.69372)
S_3	4.6049	269.24	0.0001	32→37 (0.68452) 32→39 (-0.16796)
S_4	4.8172	257.38	0.0000	35→37 (0.55910) 35→39 (0.10673) 36→38 (-0.41544)
S_5	5.5094	225.04	0.0966	35→37 (0.43003) 36→38 (0.55055)

Table 10: TD-DFT excited states of *p*-nitroaniline in water at B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	3.7110	334.10	0.4033	36→37 (0.70361)
S_2	3.9731	312.06	0.0000	34→37 (0.70046)
S_3	4.3593	284.41	0.0032	35→37 (0.67218) 36→38 (-0.20984)
S_4	4.4987	275.60	0.0001	33→37 (0.70034)
S_5	5.0294	246.52	0.0590	35→37 (0.20731) 35→39 (-0.15654) 36→38 (0.65616)

Table 11: TD-DFT excited states of *p*-nitroaniline in water at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	4.1014	302.30	0.0000	34→37 (0.68447) 34→39 (-0.17253)
S_2	4.1350	299.84	0.4664	31→37 (-0.10478) 36→37 (0.69371)
S_3	4.6079	269.07	0.0001	32→37 (0.68463) 32→39 (-0.16741)
S_4	4.8134	257.58	0.0000	35→37 (0.56136) 35→39 (0.10609) 36→38 (-0.41258)
S_5	5.5053	225.21	0.0957	35→37 (0.42712) 35→39 (-0.10017) 36→38 (0.55266)

Table 12: TD-DFT excited states of protonated *p*-nitroaniline ($pNAH^+$) in acetonitrile at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	4.1192	300.99	0.0000	34→37 (0.66699) 34→39 (-0.23096)
S_2	4.4605	277.96	0.0000	33→37 (0.66985) 33→39 (-0.22002)
S_3	4.9698	249.47	0.0305	35→38 (0.24808) 36→37 (0.65458)
S_4	5.3596	231.33	0.2712	35→37 (0.69008) 36→38 (-0.11131)
S_5	6.4385	192.57	0.0012	32→37 (0.48509) 35→38 (-0.40824) 36→37 (0.20474) 36→39 (-0.22872)

Table 13: TD-DFT excited states of protonated *p*-nitroaniline ($pNAH^+$) in water at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	4.1249	300.57	0.0000	34→37 (0.66774) 34→39 (-0.22875)
S_2	4.4649	277.68	0.0000	33→37 (0.67050) 33→39 (-0.21791)
S_3	4.9595	249.99	0.0303	35→38 (0.24578) 36→37 (0.65563)
S_4	5.3468	231.88	0.2709	35→37 (0.69021) 36→38 (-0.11010)
S_5	6.4410	192.49	0.0005	32→37 (0.46410) 35→38 (-0.42276) 36→37 (0.20882) 36→39 (-0.24255)

Table 14: TD-DFT excited states of protonated p-nitroaniline ($pNAH^+$) in benzene at CAM-B3LYP/6-31G(d).

State	Energy (eV)	Wavelength (nm)	f	Dominant contribution
S_1	3.9892	310.80	0.0000	34→37 (0.64663) 34→39 (0.28206)
S_2	4.3571	284.56	0.0000	32→37 (0.65193) 32→39 (0.26867)
S_3	5.1595	240.30	0.0314	35→38 (0.29525) 36→37 (0.62777) 36→39 (-0.10475)
S_4	5.6215	220.55	0.2612	35→37 (0.68428) 36→38 (-0.15038)
S_5	6.2285	199.06	0.0399	33→37 (0.65378) 35→38 (-0.17742) 36→37 (0.17740)

B Derivation of the Excited-State Acidity Relation

The ground-state acidity is defined as

$$pK_a = \frac{\Delta G}{2.303RT} \tag{3}$$

while the excited-state acidity is given by

$$pK_a^* = \frac{\Delta G^*}{2.303RT} \tag{4}$$

Using the following relationship,

$$\Delta G^* = \Delta G + \Delta E \tag{5}$$

where ΔE is the difference in excitation energies between the protonated (acid) and neutral (base) forms:

$$\Delta E = E_{\text{acid}} - E_{\text{base}} \tag{6}$$

Substituting into the definition of pK_a^* gives

$$pK_a^* = pK_a - \frac{\Delta E}{2.303RT} \tag{7}$$

At 298 K, $2.303RT \approx 0.0592$ eV, leading to

$$\Delta pK_a^* \approx -16.91\,\Delta E(\mathrm{eV}) \tag{8}$$

In this work, vertical excitation energies from TD-DFT calculations were used, so the resulting pK_a^* values should be interpreted as approximate trends.